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Lithium in the pre-main sequence triple system UX Tauri *

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Abstract. We present high resolution spectra (range 655–675 nm) of the pre-main sequence triple system UX Tau. Favourable seeing conditions allowed us to obtain spectra of the faintest component, where we were able to detect Hα and the Li I λ670.7 nm feature. An estimate of effective temperature, luminosity, and mass leads to the conclusion that we are dealing with a very young, very low mass object, close to the mass limit separating stars from brown dwarfs.

Lithium abundances and evolutionary status of the system are discussed, with special attention to the faint companion. A powerful observational test for evolutionary models of very low mass stars may come from lithium measurements, which can set restrictions to the maximum internal temperature attained by these objects.

Key words: stars: abundances – stars: binaries: visual – stars: late type – stars: pre-main-sequence

1. Introduction

Despite the great observational and theoretical effort devoted to the investigation of lithium depletion in main sequence low mass stars (F-M spectral types), the processes responsible for this depletion have not yet been fully understood. One of the main obstacles towards the comprehension of this problem is the lack of information on the lithium destruction rate during the pre-main sequence (PMS) evolution of these stars.

Lithium measurements in PMS stars are scarce compared with those of the main sequence, although Li in T Tauri stars has been known to exist for more than 40 years. The first quantitative analyses (Bonsack & Greenstein 1960; Zappala 1972) dealt with stars with $M > 2\,M_{\odot}$, more massive than F main sequence stars, and did not find evidence of Li destruction, in agreement with Bodenheimer's (1965) calculations for massive stars. For masses lower than $1\,M_{\odot}$ Bodenheimer found high Li depletion during PMS evolution. More recent calculations by D'Antona & Mazzitelli (1984) and Proffit & Michaud (1989) qualitatively agree with Bodenheimer's results, even though they predict slightly lower PMS Li depletion for low mass stars.

Observations in F-M stars of very young clusters, α Per (Balachandran et al. 1987), Pleiades (Duncan & Jones 1983; Butler et al. 1987; García López et al. 1990), IC 2391 (Stauffer et al. 1989), have shown stars with severe Li depletion. The ages of these clusters constrain the depletion time scale to less than $3–5\ 10^7$ yr. However, it is still not clear whether the depletion occurs during the PMS or after the stars have settled on the main sequence. Magazzù & Rebolo (1989) and Strom et al. (1989a) show the existence of strong depletion of lithium in several low mass PMS stars. The PMS stars with $0.4 \lesssim M \lesssim 0.7\,M_{\odot}$ generally present higher Li depletion than those with $M \gtrsim 0.8\,M_{\odot}$. However, for a given mass, the Li abundance presents a large scatter and does not show a clear correlation with age. We recall that large uncertainties in the estimation of masses and ages can be introduced by alterations of the position of T Tauri stars in the HR diagram due to circumstellar envelopes and disks (see e.g. Kenyon & Hartmann 1990), and by the comparison with controversial evolutionary tracks.

These uncertainties are considerably reduced in PMS multiple systems, where one can determine the age and the mass of the components with higher precision than in the remaining PMS stars. In the course of a programme of high resolution spectroscopy of visual and spectroscopic binary systems, we have found of particular interest the UX Tau system: two well known weak T Tauri stars (Strom et al. 1989a) and a third, fainter component at about 2–3″ from UX Tau A, discovered in 1975 by Herbig (UX Tau C; see Jones & Herbig 1979). Although speckle observations (Baier et al. 1985) failed to detect this faint companion, it was recently seen in near infrared images of the system by Moneti & Zinnecker (1991).

In this paper we confirm the detection of UX Tau C in the optical range, presenting its spectrum. We estimate its effective temperature and luminosity, and conclude that UX Tau C is a very low mass object close to the mass limit to ignite stable hydrogen burning. We provide Li I λ670.7 nm and Hα measurements for the three components of the system, discussing its evolutionary status with special attention to the nature of UX Tau C.

2. Observations

The spectra were obtained on the nights of March 9, 10, and 12 1990, using the 2.5 m Isaac Newton Telescope, equipped with the Intermediate Dispersion Spectrograph, the 500 nm camera, and a GEC CCD detector (pixel size 22 μm). The favourable seeing (better than 0″.7) allowed us to clearly detect UX Tau C, as shown

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* Based on observations made with the Isaac Newton Telescope, operated on the island of La Palma by the Royal Greenwich Observatory in the Spanish Observatorio del Roque de los Muchachos of the Instituto de Astrofísica de Canarias.

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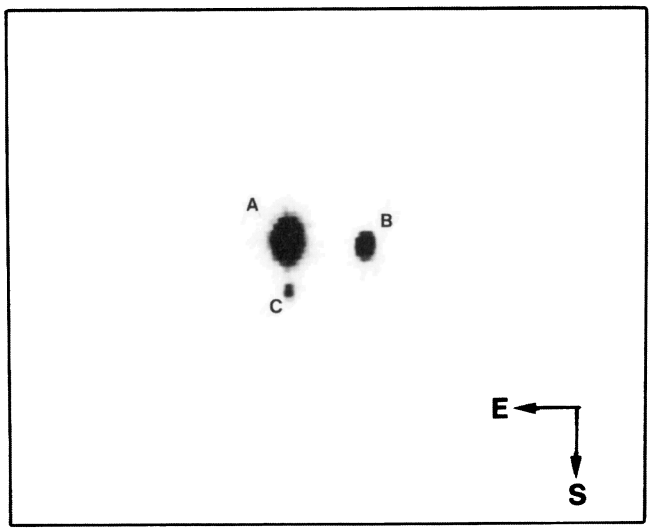


Fig. 1. An image of the UX Tau triple system taken with the guiding TV camera of the Isaac Newton Telescope

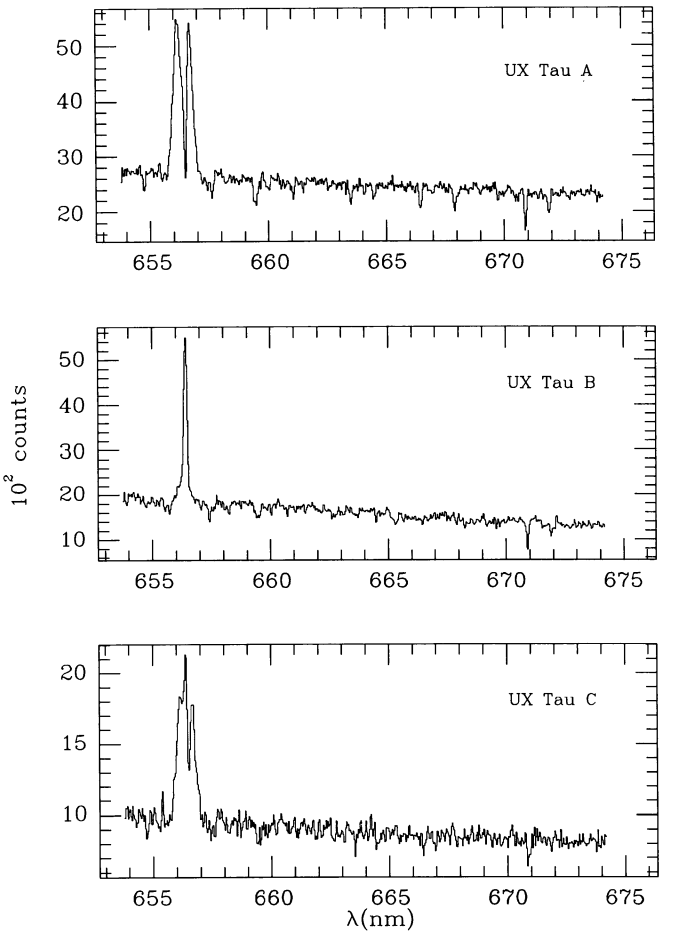


Fig. 2. Our 36 pm pixel^{-1} spectra of the UX Tau system. Note the correspondence between the side peaks of Hα in the UX Tau C spectrum and the double-peaked Hα in UX Tau A

by an image of the guiding TV camera (Fig. 1), and to take spectra of this faint component.

On the first night, we observed the UX Tau system using a grating with a reciprocal dispersion of 36 pm pixel^{-1}, which allowed us to have both the Li I λ670.7 nm line and Hα in the same range. On the night of March 10 the three stars of the system were observed only in the lithium region, with a higher resolution grating (reciprocal dispersion 22 pm pixel^{-1}). These latter observations were repeated, as a test, on the 12. No appreciable difference was found between these two sets of spectra. Exposure times ranged from 500 to 1500 s. All the observations were performed with a slit width of 150 μm (0.″8 on the sky), giving an effective spectral resolution of about 2 pixels, in either configuration. Data reduction was carried out with the IRAF package [1] at the Instituto de Astrofísica de Canarias. The CCD frames were bias corrected, flat fielded, and sky subtracted. The resulting spectra were wavelength calibrated via the emission spectrum of a CuAr lamp, recorded after each object exposure.

We re-observed the system during the revision process of this paper, namely on October 9, 1990. The same configuration as above (22 pm pixel^{-1}) was adopted, but in these new observations we were able to rotate the slit and obtain spectra of both UX Tau A and C in the same frame. This allowed us to measure any possible straylight (from UX Tau A) affecting the spectra of UX Tau C (see next section).

3. Hα and the Li I λ670.7 nm doublet

In Fig. 2 we show the spectra taken with the lower dispersion configuration. We notice that Hα is in emission in all three components. The Hα equivalent widths of UX Tau A and B (Table 1) are quite similar to those obtained from the low resolution data of Cohen & Kuhi (1979). Despite the very good seeing, some light coming from UX Tau A, the brightest component of the system, located at less than 3″ from UX Tau C, contaminated the spectrum of this faint object. In fact, two of the three peaks in the Hα of UX Tau C can be clearly attributed to the brighter companion. This allowed us to measure, with a good degree of precision, the fraction of light due to UX Tau A. In Fig. 3 we present the decontaminated spectrum of UX Tau C, showing Hα in emission with an equivalent width of 1.0 ± 0.3 nm. The error introduced by the decontamination procedure is smaller than the indetermination due to the noise in the continuum. We note that the only feature clearly visible is Hα, while any other feature lies below the detection limit imposed by the noise ($S/N \sim 1.5$, at 2σ). In particular, this renders undetectable the Li I λ670.7 nm line and the molecular bands expected in this faint component, which are clearly seen in the higher resolution spectrum.

The higher resolution spectra taken on March 12 are shown in Fig. 4. From the position of relevant lines we measured the heliocentric radial velocities of the two brightest components, $+21$ km s^{-1} for UX Tau A and $+18$ km s^{-1} for UX Tau B, with an estimated error of ± 2 km s^{-1} (rms). Slightly lower velocities, $+18$ km s^{-1} for UX Tau A and $+15.6$ for UX Tau B, were obtained previously from Herbig (1977), and Hartmann et al. (1986), respectively. The radial velocity of UX Tau C, estimated from the molecular bands, agrees within 10 km s^{-1} with that of its

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Table 1. Parameters, equivalent widths, and abundances of the UX Tau system. Figures in *italics* are upper limits

Object	SpT	L ($L_\odot$)	$T_{\rm eff}$ (K)	M ($M_\odot$)	log g	Age (10^6 yr)	W_λ(Hα) (nm)	W_λ(Li I λ670.7 nm) (pm)	log N(Li)
UX Tau A	K2	1.3	4460	1.25	4.0	3.0	0.65±0.05	43.0± 1.5	3.2
UX Tau B	M1	0.5	3500	0.4	3.5	1.0	0.35±0.05	60.0± 3.0	2.3
UX Tau C	M6:	*0.1*	*3100*	*0.2*	3.6	*1.0*	1.0 ±0.3	80.0±10.0	1.5–2.7?

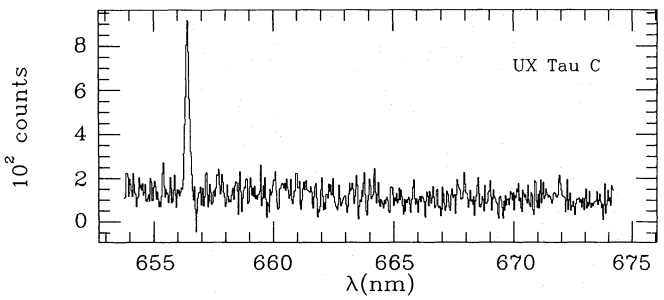


Fig. 3. The decontaminated spectrum of UX Tau C in the Hα region

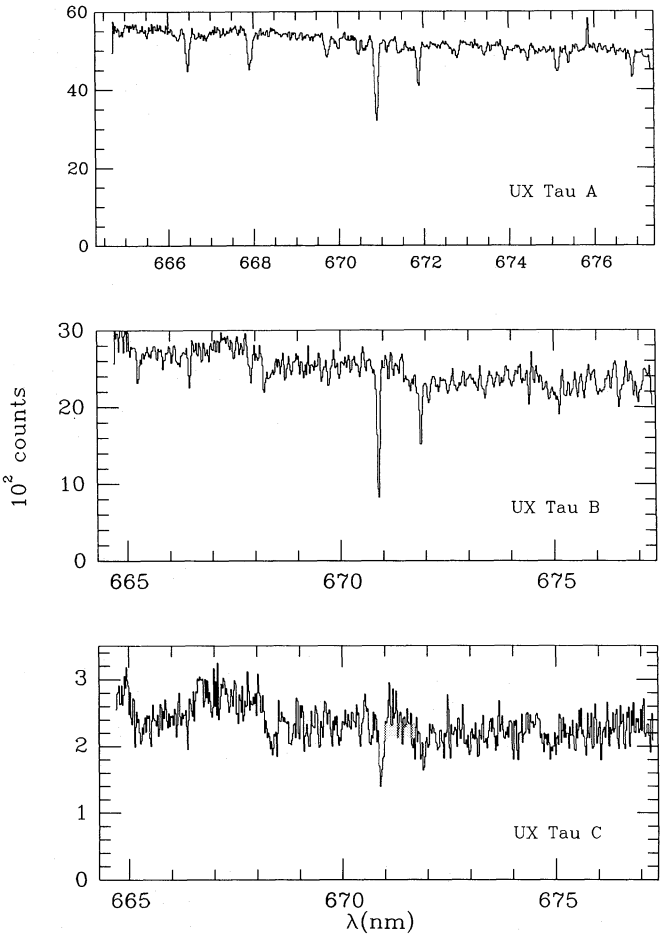


Fig. 4. The 22 pm pixel^{-1} spectra of the three components of the UX Tau system (Observations of March 1990). The Li I λ670.7 nm line can be easily seen in each spectrum

companions. The two peaks of the Hα line in UX Tau A are shifted $\pm 120\,{\rm km\,s^{-1}}$ with respect to the wavelength expected from the radial velocity of the star. The shift of the single-peaked Hα in UX Tau B and C is less than $+10\,{\rm km\,s^{-1}}$.

Since UX Tau A and B are weak T Tauri stars, their spectra can be regarded as free from veiling effects, which may considerably reduce the measured equivalent widths (for details see e.g. Hartigan et al. 1989; Basri & Batalha 1990). The equivalent widths of the Li I λ670.7 nm doublet measured in the three stars of our system are listed in Table 1. We present the weighted average of the measurements made in all the higher resolution spectra, which agree within the quoted error bar. Strom et al. (1989a) measured $W_\lambda = 35$ pm for the Li I λ670.7 nm feature in UX Tau A; the difference with our value (43 pm) is probably due to the lower resolution and S/N ratio of their data, but intrinsic variations cannot be excluded.

In the spectrum of UX Tau C shown in Fig. 4 it was not possible to assess the extent of the contamination, if present. For this reason, we repeated the higher resolution observations. On October 9 we obtained two long-slit frames of UX Tau C with its bright companion. The study of the spatial profile of each frame allowed us to derive a straightforward estimate of the contamination from UX Tau A, which turned out to be negligible. In Fig. 5 the sum of the new spectra of UX Tau C is compared with the spectra of two M dwarf stars, GL 735 (M2e) and GL 15B (M6), taken with the same configuration (unpublished data by Rebolo et al. 1985). Note the presence of the Li I λ670.7 nm line and the TiO bands in UX Tau C. The intensity of these bands indicates a spectral type close to M6. This spectral type corresponds to an effective temperature of about 3100 K, which agrees with the $T_{\rm eff}$ derived from the photometry (Sect. 4.1).

On these spectra, we repeated the measurement of the heliocentric radial velocity, obtaining $+15\pm1\,{\rm km\,s^{-1}}$ for UX Tau A and $+11\,{\rm km\,s^{-1}}$ for UX Tau C (one pixel uncertainty). The difference between $+15\,{\rm km\,s^{-1}}$ and the velocity obtained from the March observations ($+21\,{\rm km\,s^{-1}}$) is not greater than 3σ. Further observations are required to establish whether the radial velocity of UX Tau A varies or not.

4. Discussion

4.1. Evolutionary status of the system

In Table 1 we list the parameters of the stars of our system. Luminosities are taken from Strom et al. (1989b). Temperatures have been calculated from the spectral types, taken from Cohen & Kuhi (1979), according to the scale by de Jager & Nieuwenhuijzen (1987) for luminosity class IV stars. Effective temperatures for T Tauri stars are usually estimated using the Cohen & Kuhi (1979) scale (luminosity class V). However, these latter authors adopted class V instead of IV (which they considered the most appropriate for PMS stars) only because of the greater availability (at that

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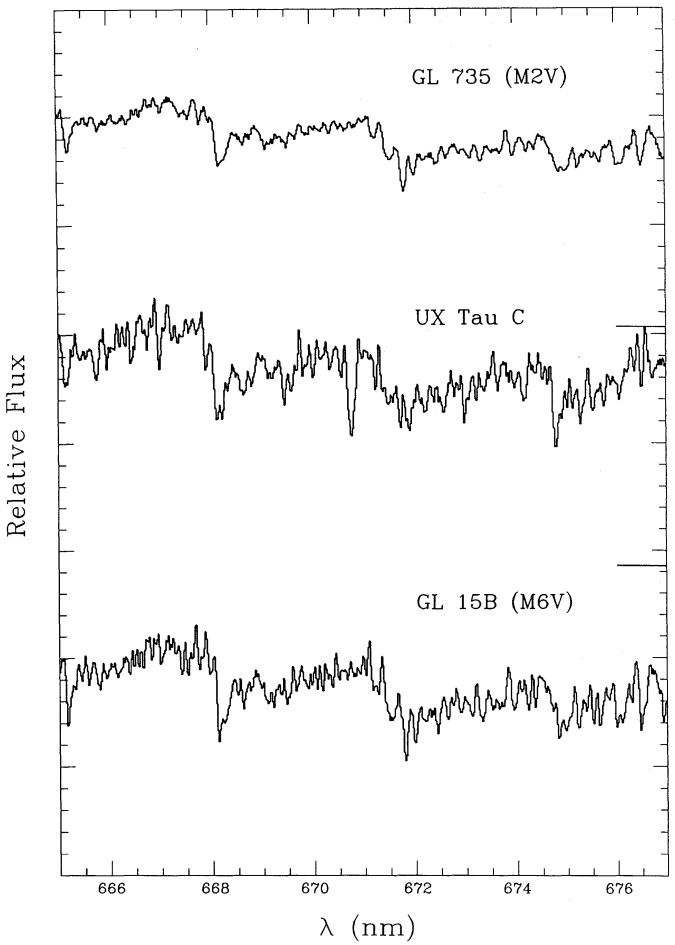


Fig. 5. The sum of the 22 pm pixel^{-1} spectra of UX Tau C taken in October 1990, together with the spectra of two M stars. The scale is the same for the three spectra; zero levels are indicated by horizontal lines. A two point boxcar smoothing has been performed in all the spectra. Note the similar intensity of the TiO bands in UX Tau C and in the M6 star GL15B

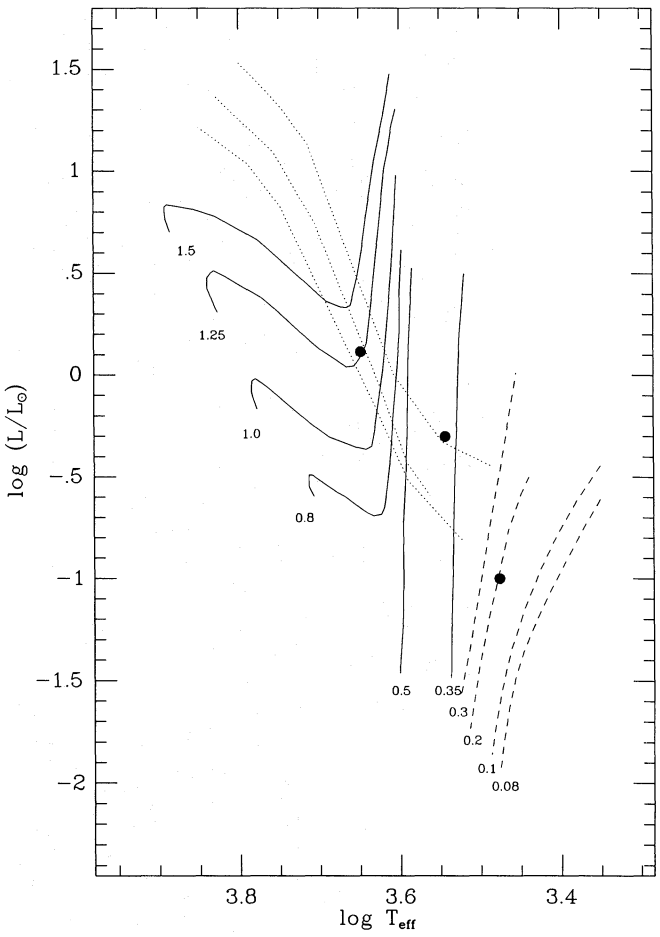


Fig. 6. HR diagram for the UX Tau system. Evolutionary tracks are from Cohen & Kuhi (1979; continuous line) and D'Antona & Mazzitelli (1985; dashed). Isochrones (dotted) are from Cohen & Kuhi (1979) and represent 1 10^6 yr (upper curve), 1.75 10^6, and 3 10^6 yr (lower curve)

time) of reliable temperature scales for the former class. Mass, gravity, and age have been derived from a comparison with evolutionary tracks and isochrones from Cohen & Kuhi (1979), shown here in Fig. 6. Taking into account the errors induced by this procedure, we note that the ages – about 3 10^6 and 1 10^6 yr for UX Tau A and B, respectively – agree with the hypothesis of coevality of the components.

At first sight one could conclude that the high IR excess of UX Tau A, discussed in Strom et al. (1989b), is mainly due to the presence of its faint companion. Recent IR imaging by Moneti & Zinnecker (1991) shows that this is not the case. In fact, they obtain for UX Tau C: $J = 11.71$, $H = 10.84$, and $K = 10.53$, which indicate that this object is too faint to contribute appreciably to the infrared excess observed in its bright companion. Strom et al. (1989b) argued that this excess could be due to emission from a circumstellar disk. However, Bertout et al. (1988), in their study on accretion disks around T Tauri stars, claimed that an optical thick, extended disk is not expected around UX Tau A. According to them, in fact, this star is likely to have already accreted most of its disk.

In order to obtain the luminosity and effective temperature of UX Tau C, we have fitted a black body curve to the above infrared data. This fairly crude method is, at present, the only one available to study stars with such low temperature; its limitations are discussed, for example, in a review by Liebert & Probst (1987). Figure 7 shows the fit with $T = 3100$ K, a coarse approximation to the photometric data, flux calibrated and dereddened according to Koornneef (1983). A value of $A_V \sim 0.5$, the average of the visual reddening of the A and B components (Strom et al. 1989b), has been adopted. The determination of the reddening is not critical in what follows, since one magnitude uncertainty in A_V allows to deredden the K flux within to $0^m.1$. We assumed a distance of 160 pc to the Taurus-Auriga star forming region (Cohen & Kuhi 1979).

The fit with $T = 3100$ K gives a luminosity $L = 0.11\, L_\odot$. The possible presence of a dust envelope, or circumstellar disk around UX Tau C has been studied in detail. We find that the photometry may be consistent with a dust component ($T \sim 2500$ K), which might contribute up to 0.7 of the total observed flux. In this case, the UX Tau C luminosity would be lowered to 0.08 (solar units). It has been claimed (Reid & Gilmore 1984) that H_2O bands lead to overestimating the luminosity because the effective wavelengths of the *JHK* filters lie between the absorption bands. This is also true for the water band at 3 μm (Berriman & Reid 1987). As a

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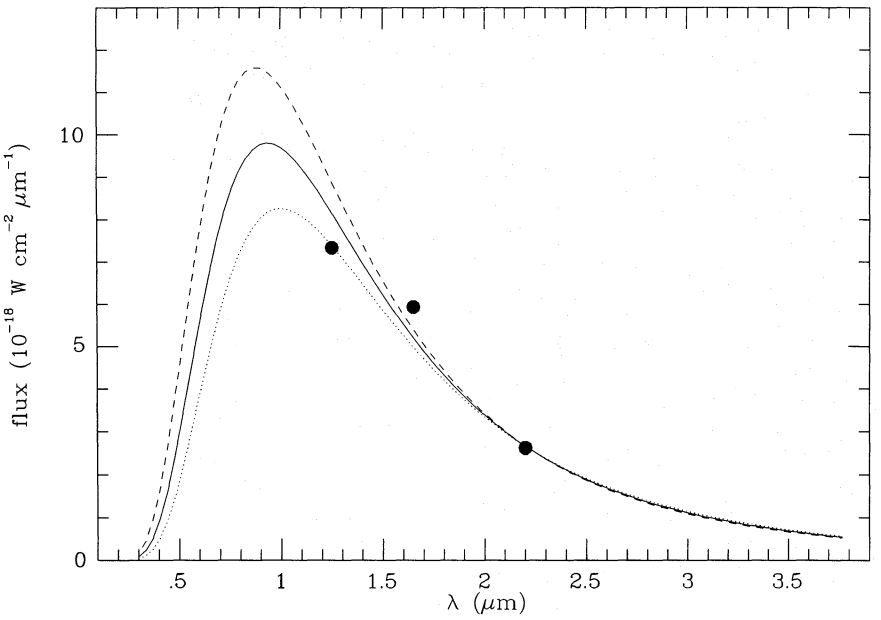


Fig. 7. Infrared data (*JHK* photometry) for UX Tau C. The three Planck curves are normalized to the *K* flux, dereddened as in the text. Temperatures are as follows: 2900 K (dotted line); 3100 K (continuous); 3300 K (dashed)

consequence, the effective temperature would also be overestimated. Berriman & Reid find that, for the coolest stars ($T_{\rm eff} \lesssim 3000$ K), $M_{\rm bol}$ can be overestimated as much as 20%, and $T_{\rm eff}$ as much as 300 K. We conclude that the parameters we find for UX Tau C have to be considered as upper limits. In particular, we have to bear in mind that a temperature as low as 2800–2700 K might be appropriate for our object.

From a comparison with the HR diagram of Fig. 6 (evolutionary tracks by D'Antona & Mazzitelli 1985), we can set an upper limit of 0.2 $M_\odot$ to the mass of UX Tau C. Due to the above uncertainties in luminosity and temperature, our faint object may have a mass near 0.08 $M_\odot$, which defines the boundary between stars and brown dwarfs (D'Antona & Mazzitelli 1985), or even lower. Further observations are required to assess whether UX Tau C is a young brown dwarf.

An estimate of the age of UX Tau C can be obtained from the luminosity vs. age curves computed by D'Antona & Mazzitelli (1985). From these calculations we can see that, whatever the mass (lower than 0.1 $M_\odot$) of an object, a luminosity around 0.08 $L_\odot$ can be obtained only for an age lower than $\sim 10^6$ yr. For UX Tau C such age is in close agreement with the age obtained for the other two components of the system (see Sect. 4.1). This gives strong support to a common origin for the three components.

4.2. Lithium abundances

4.2.1. UX Tau A and B

We derived Li abundances with a modified version of the SYNTHE code (Castelli 1987), using models provided by Dr. Gustafsson, calculated using the same assumptions as Bell et al. (1975). The adopted stellar parameters are those listed in Table 1. A microturbulence $\xi = 2.0$ km s^{-1} has been used. A recent detailed curve of growth analysis of Sz19, a hotter (G2) T Tauri star, led to a value of $\xi = 1.6$ km s^{-1} (Franchini et al. 1991), which is not very far from our choice. In any case, we estimate that an uncertainty of ± 1 km s^{-1} in microturbulence introduces an error of less than ± 0.15 dex in the abundances. Assuming that the equivalent width of the Li I λ670.7 nm doublet is entirely due to ^{7}Li, we obtain $\log N(\mathrm{Li}) = 3.2$ and 2.3 (in the scale $\log N(\mathrm{H}) = 12$) for UX Tau A and B, respectively, with an estimated total uncertainty of about ± 0.35 dex.

Since the luminosity class of T Tauri stars remains a matter of debate, the choice of the $T_{\rm eff}$ calibration scale may add uncertainty in the absolute values of the Li abundance. In the case of UX Tau A, from the temperature scale of Cohen & Kuhi (1979) one would assign a temperature about 500 K higher than the $T_{\rm eff}$ used in this work, which would considerably increase the derived lithium abundance. Basri et al. (1990) point out that the Li abundance in this object would lie in the range $\log N(\mathrm{Li}) = 4.2$–$3.0$, depending on the choice of $T_{\rm eff}$ and ξ. Adopting the calibration scale by de Jager & Nieuwenhuijzen (1987) for luminosity class IV stars, we obtain a Li abundance for UX Tau A very similar to the abundance derived in meteorites and in many young open cluster stars (see references in Rebolo 1990). This agrees with predictions of no appreciable depletion in solar mass stars during their PMS phase (Bodenheimer 1965; D'Antona & Mazzitelli 1984).

We see that UX Tau B presents a Li depletion of almost one order of magnitude with respect to UX Tau A. In a 0.4 $M_\odot$ star no Li depletion should occur before 10^7 yr, and a depletion of one order of magnitude should be observable at an age of $\sim 2\ 10^7$ yr (see Table 1 of Strom et al. 1989a). This leads to an inconsistency between age and lithium abundance in UX Tau B. This abundance has been derived with a curve of growth obtained from the extrapolation of curves from slightly hotter (~ 500 K) models, and we cannot exclude that such extrapolation is the cause of part of the observed discrepancy.

We note also that non-LTE effects in the formation of the Li I λ670.7 nm resonance line in late type stars may rise the lithium abundances computed in LTE, as outlined by Steenbock & Holweger (1984) and, more recently, by Pavlenko (1989, 1990). These authors consider stars with lower gravity than in T Tauri stars, but with similar $T_{\rm eff}$. Although the higher gravity would tend to reduce non-LTE effects, the higher ultraviolet radiation field present in most T Tauri stars would work in the opposite sense. Accurate model atmospheres for cool stars are needed to establish which is the dominant process and, consequently, whether non-LTE effects in PMS stars are important.

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In addition to that, one should also take into account departures from equilibrium. In fact, the T Tauri characteristics indicate that equilibrium models may be quite far from describing the atmosphere of a PMS star. The low Hα emission and scarce UV excess in both UX Tau A and B (photometry by Rydgren et al. 1982) suggest that in our case the effects of lack of equilibrium are relatively small. To check this point, we want to verify whether the models used in this work are able to derive a reasonable Fe abundance in our program stars. In the case of UX Tau A, we considered two lines of Fe I (λ666.35 and 670.51 nm), which reproduced the solar Fe abundance ($\log N(\mathrm{Fe}) = 7.5$), i.e. the normal abundance in very young open clusters (Boesgaard & Friel 1990). This result supports the hypothesis that the departure from equilibrium does not affect in a sensible way the derived abundances. For UX Tau B, due to the presence of bands and the lower S/N of the spectra, the only useful line was the one at λ666.35 nm, which returned a Fe abundance about 0.6 dex below the solar value. However, this low-excitation transition has been found to be affected from non-LTE effects in cool giants (García López et al. 1988). Although we have just seen that these effects are negligible in UX Tau A, they might be significant in UX Tau B, due to its lower gravity. In conclusion, our test, while successful in UX Tau A, is not applicable to UX Tau B. Therefore, the lithium abundance we derived for the latter star has to be taken with some additional caution.

4.2.2. UX Tau C

Our lithium detection in UX Tau C is particularly interesting. We have seen in the previous section that the mass of $0.2\,M_\odot$ is an upper limit for this object; the true mass would probably range between 0.1 and $0.08\,M_\odot$. If we assume that UX Tau C is coeval with the other two components of the system, we see that the evolutionary models by D'Antona & Mazzitelli (1985) predict that the central temperature of our object must be lower than the minimum needed to destroy lithium ($2.4\ 10^6$ K), whatever its mass (lower than $0.1\,M_\odot$). New models by Pozio et al. (1991) confirm that, even if we assign to UX Tau C our upper limits for luminosity and $T_{\rm eff}$, lithium burning in this object should have not started yet. The Li I λ670.7 nm equivalent width measured in UX Tau C, when translated to abundance, would provide a powerful observational test for such predictions.

Unfortunately, satisfactory model atmospheres for very low mass stars are not yet available (see Liebert & Probst 1987), due to the difficulties posed by the formation of molecules and their complicated opacities. We tried to obtain a qualitative indication of the Li abundance in UX Tau C by using a model with $T = 3000$ K, $\log g = 4.75$ by Mould (1976). We know that, as stressed by the author, these models are not meant for a detailed spectrum synthesis. Moreover, the synthesis code we used does not take into account the opacity sources that could be relevant at such a low temperature. The abundance we obtain, after an upward correction taking into account the lower gravity of the object with respect to the model, lies in the range $\log N(\mathrm{Li}) = 1.5$–2.2. It is worth noting that an extrapolation to $T = 3000$ K of hotter Gustafsson's models (with $\log g = 3.5$) leads to an abundance between 2.3 and 2.7. These results, although affected by large uncertainties due to the neglect of the proper opacities, both in the models and in the synthesis code, may be compatible with no significant Li depletion in UX Tau C, which would imply that the predictions by D'Antona & Mazzitelli (1985) and Pozio et al. (1991) are essentially correct. However, we cannot establish definitely whether the initial lithium abundance in UX Tau C has been preserved.

4.3. *Lithium in very low mass objects*

The determination of mass for very low mass (VLM) objects is quite hard and may be affected by large uncertainties. Despite the progress made in recent years, it is still very difficult to assess whether a brown dwarf candidate is really a substellar object on the basis of its estimated mass. For example, the mass of GL 473AB ($M = 0.07\,M_\odot$) locates this object just below the hydrogen burning mass limit ($0.08\,M_\odot$), but the $0.02\,M_\odot$ indetermination quoted by Liebert & Probst (1987) makes it impossible, at present, to draw any conclusion on its nature.

The effective temperatures and luminosities of the objects close to the hydrogen burning limit are in the range 2700–3100 K and around $10^{-3}\,L_\odot$, respectively (Burrows et al. 1989). Our detection of lithium in UX Tau C, whose temperature lies in the same range, confirms that lithium, if present, can be successfully detected in VLM objects. Theoretical predictions (D'Antona & Mazzitelli 1985; Burrows et al. 1989) indicate that the central temperature of objects with mass lower than $0.06\,M_\odot$ never reaches the value needed to destroy lithium. If these calculations are correct, we expect to find no appreciable lithium destruction in these objects.

These considerations suggest that lithium can be used successfully in the search for brown dwarfs. In fact, the detection of lithium in VLM stars and a measure of its abundance – possible when reliable model atmospheres become available – would provide a criterion to establish the mass and the nature of these objects. This would also provide a clue to set restrictions to their maximum central temperature.

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Note added in proof: Recently, Magazzù, Rebolo, and Pavlenko (paper submitted to ApJ) have studied the NLTE effects in the formation of the lithium resonance doublet in cool T Tauri stars. Their NLTE corrections applied to UX Tau A and B lead to lithium abundances $\log N(\mathrm{Li}) = 3.1$ and 3.2, respectively. This result shows that no depletion has occurred in UX Tau B, removing the inconsistency issued in Sect. 4.2.1.

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